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SQL Aggregate functions & Operations

01

To list all the departments from the Student table.

SELECT DISTINCT department

From Student;

Department
IT
HR
Finance

02 Get the average age of Student (~~from the~~ Per the) department

SELECT department; AVG (age) As avg_ age.

from Students

Group By department;

Department	avg_ age.
Finance	23
HR	22
IT	20

03 Show departments with more than 1 Student.

SELECT department, Count (*) As Student_Count

From Student

Group By department

Havin Count (*) > 1;

Department	Student
HR	2
It	2

04

Get all Students whose age is between 21 and 23.

```
SELECT Student_Id,
       Name;
       age;
       department
From Student
WHERE age BETWEEN 21 AND23.
```

Student_Id	Name	age	Department
2	BOB	22	HR
3	charle.	21	IT
4	Diana	23	finance
5.	EVE	22	HR

05

LIST all Students in the IT or HR departments who are older than 21

```
SELECT Student_Id;
       Name;
       age;
       department
From. Students
WHERE DERAMENT IN ('IT' OR 'HR')
AND age 721
```

Student_Id	Name.	age.	DEPARTMENT
2	BOB	22	HR
5	EVE	22.	HR

02 THE COURCES TABLE.

06

Show total Credits per department, only for departments with more than 5 total credits

SELECT department;

Sum (credits) AS total_credits

From Courses,

Group By department

Having Sum (credits) > 5;

department	total_Credit
IT	11

07 list all courses that do not have Credits.

SELECT Course_Id;

Course_Name;

department;

credits

From courses

WHERE Credits <> 4;

course Id	Course_name	deparment	Credits.
101	SQL Basic	IT	3
104	Excel	finance	2
105.	Statistics	HR	3

08

Show the top 3 Courses by credits in descending order.

SELECT course_Id;

course_Name;

credits

From Courses

ORDER BY Credits DESC

LIMIT 3;

Course_id	Cours_name	Credits
102	Python	4
103	Data Science.	4
101	SQL Basic	3

Q3 The enrollment Table.

Q9 Get the Maximum, Minimum and average grade. across all enrollments.

```
SELECT MAX (grade) AS Max_grade;
       Min (grade) AS min_grade;
       AVG (grade) AS avg_grade.
From enrollments;
```

Max_grade	Min_grade	Avg_grade.
90	78	84.6.

10

Count how many enrollments exits per course

```
SELECT course_id;
       Count(*) As enrollment_Count
From enrollments.
Group By Course_id;
```

Course_id	Enrollment
101	1
102	1
103	1
104	1
105	1

Q4 The Salaries Table.

11 Find total Salary and total bonus per departement.

```
SELECT department;
       Sum(Salary) AS total_Salary;
       Sum (bonus) AS total_bonus
From salaries
Group By department;
```

department	total_Salary	total_bonus
finance.	70 000	600
HR	10 9000	7500
IT	12 2000	10500

12

Show departments where average Salary is above 55 000.

```
SELECT department;
       AVG (Salary) As avg_ Salary
From Salaries.
Group By department
Having AVG (Salary) > 55 000;
```

department	avg_Salary
finance	70 000
IT	61000.

13

List Employees whose Salary plus bonus is greater than 60 000.

```
SELECT employee_ id;
         name;
         Salary;
         bonus;
       (Salary + bonus) As total_ Compensation
from Salaries
WHERE (Salary + bonu) > 60 000;
```

Employee_id	name	Salary	bonus	total_Compensation
1	Tom	60 000	5000	65000
3	Spike	70 000	6000	76000
4	Tyke	62000	55 00	67500

05 The Project Table

14 Show total and average budget per department. Only including departments with average budget above 70 000,

```
SELECT department;
       Sum (budget ) As total_ budget;
       AVG (budget) As avg_ budget
from protecs
Group By department
Having AVG (budget)> 70 000;
```

department	total_ budget	AVG_ budget
IT	270 000	135 000
Finance.	80 000	80 000

15

list all project with budget between 50 000 and 120.000 excluding the Marketing deparment.

```
SELECT Project_ Id;
       Project_ name;
       department;
       budget
from projects
WHERE budget BETWEEN 50000 AND 12000
 AND department <> 'Marketing';
```

Project_ Id	Project_ name	department	budget
1	AI App	IT	120000
2	Pay Roll System	Finance	80 000
5	HR Portal	HR	50 00.